

PROJECT_FINAL_NOTEBOOK_V2_runnotes

June 10, 2026

```
[4]: %run PROJECT_FINAL_NOTEBOOK_V2.py
```

All packages imported successfully.

census_tracts :

https://www2.census.gov/geo/tiger/GENZ2023/shp/cb_2023_06_tract_500k.zip

acs_api : <https://api.census.gov/data/2022/acs/acs5/profile>

lodes_wac : <https://lehd.ces.census.gov/data/lodes/LODES8/ca/wac/>

sandag_poe :

https://gis.sandag.org/FreightViewer/data/poe_points_update.json

sandag_roads :

https://gis.sandag.org/FreightViewer/data/MajorRoads_20171002.json

sd_zoning_base :

https://geo.sandag.org/server/rest/directories/downloads/Zoning_Base_SD.geojson

Setup complete.

CRS: EPSG:3310

Buffers (m): {'buffer_quarter_mi': 402, 'buffer_half_mi': 805, 'buffer_one_mi': 1609}

LODES year: 2021

Loaded 2892 rows | Columns: ['business_acctnum', 'dba_name', 'ownership_type', 'address', 'city', 'state', 'zip', 'naics', 'activity_desc', 'industry_group', 'definition', 'tract_geoid', 'planning_area', 'full_address', 'matched_address', 'geocode_score', 'longitude', 'latitude']

Valid coordinates: 2892

GeoDataFrame CRS: EPSG:3310

Planning districts: 61 features

Clipped: 2892 -> 2500 businesses within SD planning areas

Subarea counts:

subarea

Other 1380

Otay Mesa 493

Sorrento Valley 369

Kearny Mesa 206

Miramar 52

Name: count, dtype: int64

Otay Mesa: 493 | Non-border total: 2007

Downloading LODES WAC:

https://lehd.ces.census.gov/data/lodes/LODES8/ca/wac/ca_wac_S000_JT00_2021.csv.gz
 (~30-60 seconds - CA WAC file is ~50MB compressed)
 Loaded: 252,472 CA census blocks
 San Diego County blocks: 15,976 | Unique tracts: 735
 All expected NAICS columns present.

Tract-level LODES: 735 tracts

	lodes_mfg_jobs	lodes_whole_jobs	lodes_trans_jobs \
count	735.00	735.00	735.00
mean	155.76	60.79	42.84
std	829.16	282.56	249.19
min	0.00	0.00	0.00
25%	0.00	0.00	0.00
50%	3.00	6.00	3.00
75%	19.50	22.00	13.00
max	8889.00	3798.00	4504.00

	lodes_industrial_jobs	lodes_industrial_share
count	735.00	735.00
mean	259.39	0.07
std	1211.83	0.12
min	0.00	0.00
25%	5.00	0.01
50%	21.00	0.03
75%	75.00	0.08
max	14045.00	0.83

Loading Census TIGER 2023 tracts...

Saved: lodes_enrichment_by_tract.csv
 736 rows | 7 columns

Top 5 tracts by industrial jobs:

GEOID	lodes_industrial_jobs	lodes_industrial_share
06073008350	14045.0	0.285131
06073008511	13970.0	0.130564
06073017049	11035.0	0.522565
06073017071	10323.0	0.508322
06073008346	9506.0	0.259776

Fetching SANDAG POE points...

POE points: 10 features

Distance to Nearest POE (miles) by Subarea						
	n	mean	median	std	min	max
subarea						
Kearny Mesa	206	20.545	20.745	0.667	18.278	21.953
Miramar	52	26.048	25.994	0.711	24.905	27.520

Otay Mesa	493	1.397	1.094	1.091	0.117	4.790
Other	1380	18.615	17.547	6.915	0.124	37.381
Sorrento Valley	369	25.364	24.992	0.798	24.174	27.755

Fetching SANDAG Major Roads...

Major roads: 45 features | CRS: EPSG:3310

Distance to Freight Road (miles) by Subarea

	n	mean	median	min	max
subarea					
Kearny Mesa	206	0.308	0.270	0.021	0.936
Miramar	52	0.997	0.541	0.118	3.154
Otay Mesa	493	0.321	0.275	0.006	1.189
Other	1380	0.696	0.450	0.020	3.587
Sorrento Valley	369	1.599	1.725	0.041	2.789
402m (0.25 mi):	733/2500	inside	(29.3%)		
805m (0.50 mi):	1374/2500	inside	(55.0%)		
1609m (1.00 mi):	1838/2500	inside	(73.5%)		

Buffer overlap by subarea

0.25 mi:

	n	n_inside	pct_inside
subarea			
Kearny Mesa	206	85	41.3
Miramar	52	13	25.0
Otay Mesa	493	200	40.6
Other	1380	423	30.7
Sorrento Valley	369	12	3.3

0.50 mi:

	n	n_inside	pct_inside
subarea			
Kearny Mesa	206	186	90.3
Miramar	52	25	48.1
Otay Mesa	493	403	81.7
Other	1380	733	53.1
Sorrento Valley	369	27	7.3

1.00 mi:

	n	n_inside	pct_inside
subarea			
Kearny Mesa	206	206	100.0
Miramar	52	34	65.4
Otay Mesa	493	486	98.6
Other	1380	1047	75.9
Sorrento Valley	369	65	17.6

Saved: joseph_indicators_output.csv

Saved: joseph_indicators_output.geojson

2,500 rows | 25 columns

FINAL VALIDATION SUMMARY

	dist_nearest_poe_mi		buffer_quarter_mi		
	mean	count	mean	count	\
subarea					
Kearny Mesa	20.545	206	0.413	206	
Miramar	26.048	52	0.250	52	
Otay Mesa	1.397	493	0.406	493	
Other	18.615	1380	0.307	1380	
Sorrento Valley	25.364	369	0.033	369	

	buffer_half_mi		buffer_one_mi		
	mean	count	mean	count	
subarea					
Kearny Mesa	0.903	206	1.000	206	
Miramar	0.481	52	0.654	52	
Otay Mesa	0.817	493	0.986	493	
Other	0.531	1380	0.759	1380	
Sorrento Valley	0.073	369	0.176	369	

Using in-memory pipeline output from section 8.0

Business points loaded: 2500

Columns: ['business_acctnum', 'dba_name', 'industry_group', 'naics', 'address', 'city', 'state', 'zip', 'latitude', 'longitude', 'CPNAME', 'subarea', 'dist_nearest_poe_m', 'dist_nearest_poe_mi', 'dist_san_ysidro_m', 'dist_san_ysidro_mi', 'dist_otay_poe_m', 'dist_otay_poe_mi', 'nearest_poe_name', 'dist_to_freight_road_m', 'dist_to_freight_road_mi', 'distance_category', 'buffer_quarter_mi', 'buffer_half_mi', 'buffer_one_mi', 'geometry']

=== Business count by subarea ===

subarea	
Other	1380
Otay Mesa	493
Sorrento Valley	369
Kearny Mesa	206
Miramar	52

Name: count, dtype: int64

=== Industry group by subarea ===

industry_group	Industrial Machinery Repair	Manufacturing	\
subarea			
Kearny Mesa	0	67	
Miramar	2	17	
Otay Mesa	2	126	
Other	12	505	
Sorrento Valley	1	156	

industry_group	Transportation and Warehousing	Wholesale Trade
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subarea		
Kearny Mesa	40	99
Miramar	11	22
Otay Mesa	160	205
Other	352	511
Sorrento Valley	85	127

=== Distance to nearest POE (miles) by subarea ===

	n	mean	median	std	min	max
subarea						
Kearny Mesa	206	20.545	20.745	0.667	18.278	21.953
Miramar	52	26.048	25.994	0.711	24.905	27.520
Otay Mesa	493	1.397	1.094	1.091	0.117	4.790
Other	1380	18.615	17.547	6.915	0.124	37.381
Sorrento Valley	369	25.364	24.992	0.798	24.174	27.755

=== Freight corridor buffer overlap (% inside) by subarea ===

buffer_quarter_mi:

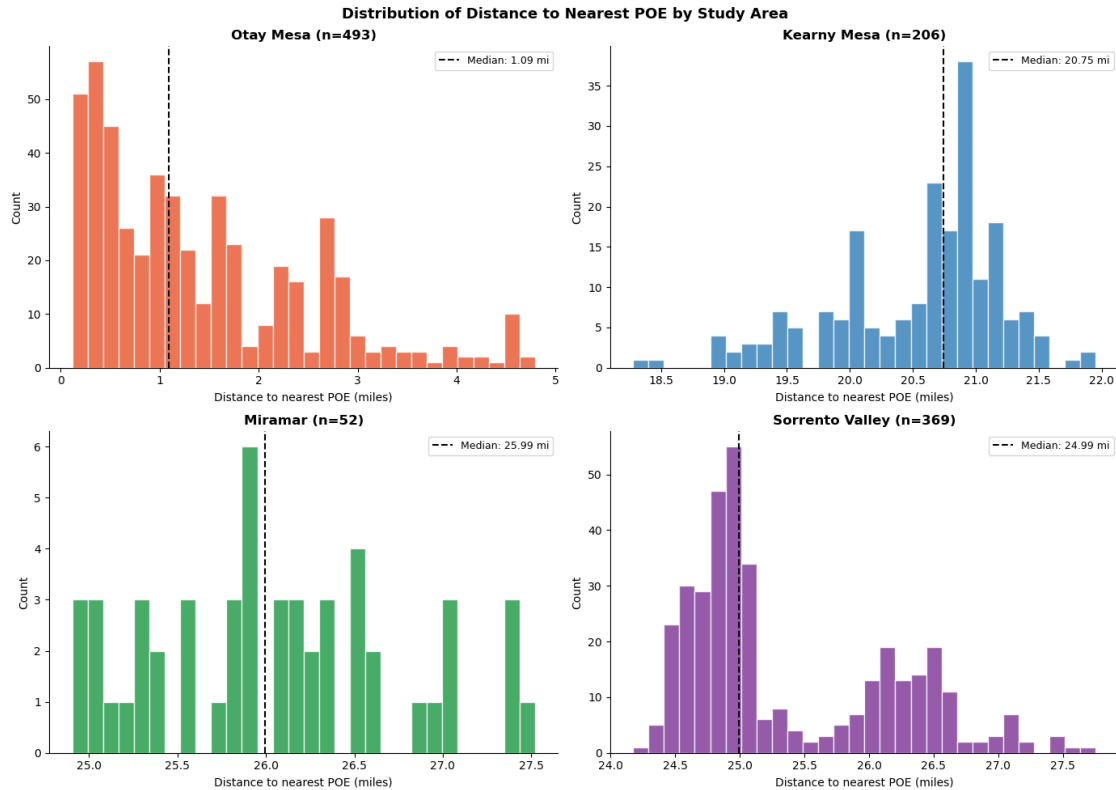
	n	n_inside	pct_inside
subarea			
Kearny Mesa	206	85	41.3
Miramar	52	13	25.0
Otay Mesa	493	200	40.6
Other	1380	423	30.7
Sorrento Valley	369	12	3.3

buffer_half_mi:

	n	n_inside	pct_inside
subarea			
Kearny Mesa	206	186	90.3
Miramar	52	25	48.1
Otay Mesa	493	403	81.7
Other	1380	733	53.1
Sorrento Valley	369	27	7.3

buffer_one_mi:

	n	n_inside	pct_inside
subarea			
Kearny Mesa	206	206	100.0
Miramar	52	34	65.4
Otay Mesa	493	486	98.6
Other	1380	1047	75.9
Sorrento Valley	369	65	17.6



Enter password:

Broad businesses: 4293

industry_group	
Transportation and Warehousing	1920
Wholesale Trade	1277
Manufacturing	1069
Industrial Machinery Repair	27
Name: count, dtype: int64	

Core businesses: 3076

industry_group	
Wholesale Trade	1188
Manufacturing	1069
Transportation and Warehousing	792
Industrial Machinery Repair	27
Name: count, dtype: int64	

San Diego County census tracts

Rows: 736

CRS: EPSG:3310

Columns: ['STATEFP', 'COUNTYFP', 'TRACTCE', 'GEOIDFQ', 'GEOID', 'NAME', 'NAMELSAD', 'STUSPS', 'NAMELSADCO', 'STATE_NAME', 'LSAD', 'ALAND', 'AWATER',

'geometry', 'area_sqmi']

	STATEFP	COUNTYFP	TRACTCE	GEOIDFQ	GEOID	NAME \
131	06	073	003212	1400000US06073003212	06073003212	32.12
132	06	073	013104	1400000US06073013104	06073013104	131.04
133	06	073	020902	1400000US06073020902	06073020902	209.02
134	06	073	015902	1400000US06073015902	06073015902	159.02
135	06	073	004501	1400000US06073004501	06073004501	45.01

	NAMLSAD	STUSPS	NAMLSADCO	STATE_NAME	LSAD	ALAND \
131	Census Tract	32.12	CA	San Diego County	California CT	1257736
132	Census Tract	131.04	CA	San Diego County	California CT	977893
133	Census Tract	209.02	CA	San Diego County	California CT	422787620
134	Census Tract	159.02	CA	San Diego County	California CT	1106949
135	Census Tract	45.01	CA	San Diego County	California CT	327790

	AWATER	geometry	area_sqmi
131	0	POLYGON ((276007.796 -589973.492, 276107.605 -...	0.491867
132	0	POLYGON ((274057.664 -595209.757, 274440.207 -...	0.377675
133	7987272	POLYGON ((300577.682 -562264.415, 300499.819 -...	166.242285
134	0	POLYGON ((283358.104 -576469.74, 283714.904 -5...	0.428744
135	0	POLYGON ((268224.337 -584056.065, 268432.335 -...	0.126785

GADM San Diego County boundary

Rows: 1

CRS: EPSG:3310

Columns: ['GID_2', 'GID_0', 'COUNTRY', 'GID_1', 'NAME_1', 'NL_NAME_1', 'NAME_2', 'VARNAME_2', 'NL_NAME_2', 'TYPE_2', 'ENGTYPE_2', 'CC_2', 'HASC_2', 'geometry']

	GID_2	GID_0	COUNTRY	GID_1	NAME_1	NL_NAME_1	NAME_2 \
219	USA.5.37_1	USA	UnitedStates	USA.5_1	California	NA	SanDiego

	VARNAME_2	NL_NAME_2	TYPE_2	ENGTYPE_2	CC_2	HASC_2 \
219	NA	NA	County	County	NA	US.CA.SD

	geometry
219	MULTIPOLYGON (((259551.989 -578069.34, 259508...

Community Planning District Boundaries

Rows: 61

CRS: EPSG:3310

Columns: ['OBJECTID', 'CPCODE', 'CPNAME', 'ACREAGE', 'Website', 'Shape_Length', 'Shape_Area', 'geometry']

	OBJECTID	CPCODE	CPNAME	ACREAGE \
0	1	97	MILITARY FACILITIES	22605.531046
1	2	7	EAST ELLIOTT	2806.769212
2	3	40	TORREY PINES	2722.328674
3	4	28	NORTH PARK	2253.719395
4	5	34	SCRIPPS MIRAMAR RANCH	4197.429037

	Website	Shape_Length	\
0	None	182376.005367	
1	https://www.sandiego.gov/planning/community/pr...	68776.912157	
2	https://www.sandiego.gov/planning/community/pr...	101824.664393	
3	https://www.sandiego.gov/planning/community/pr...	50884.200918	
4	https://www.sandiego.gov/planning/community/pr...	93586.943862	

	Shape_Area	geometry
0	9.846930e+08	POLYGON ((280399.232 -566493.238, 280574.631 -...
1	1.222624e+08	MULTIPOLYGON (((280663.22 -570149.209, 280657...
2	1.185842e+08	POLYGON ((257776.762 -557737.89, 257772.861 -5...
3	9.817162e+07	POLYGON ((267682.378 -581665.26, 267751.895 -5...
4	1.828393e+08	POLYGON ((275072.095 -561583.76, 275084.157 -5...

['BALBOA PARK' 'BARRIO LOGAN' 'BLACK MOUNTAIN RANCH'
'CARMEL MOUNTAIN RANCH' 'CARMEL VALLEY' 'CLAIREMONT MESA' 'COLLEGE AREA'
'DEL MAR MESA' 'DOWNTOWN' 'EAST ELLIOTT' 'ENCANTO NEIGHBORHOODS'
'FAIRBANKS RANCH COUNTRY CLUB' 'GREATER GOLDEN HILL' 'KEARNY MESA'
'LA JOLLA' 'LINDA VISTA' 'LOS PENASQUITOS CANYON' 'MID-CITY:CITY HEIGHTS'
'MID-CITY:EASTERN AREA' 'MID-CITY:KENSINGTON-TALMADGE'
'MID-CITY:NORMAL HEIGHTS' 'MIDWAY-PACIFIC HIGHWAY' 'MILITARY FACILITIES'
'MIRA MESA' 'MIRAMAR RANCH NORTH' 'MISSION BAY PARK' 'MISSION BEACH'
'MISSION VALLEY' 'NAVAJO' 'NCFUA SUBAREA II' 'NORTH PARK' 'OCEAN BEACH'
'OLD TOWN SAN DIEGO' 'OTAY MESA' 'OTAY MESA-NESTOR' 'PACIFIC BEACH'
'PACIFIC HIGHLANDS RANCH' 'PENINSULA' 'RANCHO BERNARDO'
'RANCHO ENCANTADA' 'RANCHO PENASQUITOS'
'RESERVE AREA-Not a community plan' 'SABRE SPRINGS' 'SAN PASQUAL'
'SAN YSIDRO' 'SCRIPPS MIRAMAR RANCH' 'SERRA MESA'
'SKYLINE-PARADISE HILLS' 'SOUTHEASTERN SAN DIEGO' 'TIERRASANTA'
'TIJUANA RIVER VALLEY' 'TORREY HIGHLANDS' 'TORREY HILLS' 'TORREY PINES'
'UNIVERSITY' 'UPTOWN' 'VIA DE LA VALLE']

Successfully geocoded before county filter: 3067

Businesses inside San Diego County census tracts: 2892

planning_area	
OTAY MESA	406
Outside City Planning Areas	392
MIRA MESA	369
KEARNY MESA	206
CLAIREMONT MESA	108
NAVAJO	96
OTAY MESA-NESTOR	87
UNIVERSITY	78
DOWNTOWN	74
MID-CITY:CITY HEIGHTS	68
PENINSULA	62
SOUTHEASTERN SAN DIEGO	61
SAN YSIDRO	52
LA JOLLA	49

PACIFIC BEACH	44
RANCHO BERNARDO	44
MIDWAY-PACIFIC HIGHWAY	44
UPTOWN	43
MISSION VALLEY	42
NORTH PARK	42

Name: count, dtype: int64

Exported: industrial_businesses_core_geocoded.csv

Final row count: 2892

industry_group	
Wholesale Trade	1118
Manufacturing	1001
Transportation and Warehousing	749
Industrial Machinery Repair	24

Name: count, dtype: int64

Successfully geocoded before county filter: 4280

Businesses inside San Diego County census tracts: 4082

planning_area

Outside City Planning Areas	595
OTAY MESA	435
MIRA MESA	430
KEARNY MESA	217
MID-CITY:CITY HEIGHTS	200
CLAIREMONT MESA	156
NAVAJO	128
OTAY MESA-NESTOR	115
DOWNTOWN	113
UNIVERSITY	101
MID-CITY:EASTERN AREA	97
SOUTHEASTERN SAN DIEGO	94
PENINSULA	87
NORTH PARK	81
MISSION VALLEY	81
ENCANTO NEIGHBORHOODS	77
SKYLINE-PARADISE HILLS	73
SAN YSIDRO	70
LINDA VISTA	67
RANCHO PENASQUITOS	65

Name: count, dtype: int64

Exported: industrial_businesses_broad_geocoded.csv

Final row count: 4082

industry_group	
Transportation and Warehousing	1841
Wholesale Trade	1204
Manufacturing	1010

Industrial Machinery Repair 27
 Name: count, dtype: int64

SANDAG major freight roads

Rows: 45

CRS: EPSG:3310

Columns: ['OBJECTID_1', 'OBJECTID', 'IFC', 'LABEL', 'EXISTING', 'Shape_Leng', 'Shape_Le_1', 'NM', 'Source', 'Mileage', 'geometry']

	OBJECTID_1	OBJECTID	IFC	LABEL	EXISTING	Shape_Leng	Shape_Le_1	\
0	5	1	1	15	1	109.350581	577371.069707	
1	6	2	2	76	1	51.965596	274378.347456	
2	10	3	2	79	1	45.528953	240392.871618	
3	15	4	2	78	1	0.015452	81.586014	
4	23	5	2	79	1	7.050024	37224.128392	

		NM	Source	Mileage	\
0	Interstate 15	SANDAG	109.3		
1	State Route 76	SANDAG	52.0		
2	State Route 79	SANDAG	45.5		
3	State Route 78	SANDAG	0.0		
4	State Route 79	SANDAG	7.0		

	geometry
0	MULTILINESTRING ((270549.256 -587154.978, 2705...
1	MULTILINESTRING ((306937.193 -529563.399, 3069...
2	MULTILINESTRING ((306935.947 -529557.886, 3069...
3	LINESTRING (273240.237 -538404.408, 273263.449...
4	LINESTRING (306937.193 -529563.399, 306944.777...

Exported: industrial_businesses_core_distance_to_freight_roads.csv

Overall summary:

	definition	total_businesses	mean_distance_mi	median_distance_mi	\
0	Core	2892	0.752046	0.450067	

	min_distance_mi	max_distance_mi
0	0.00164	4.753636

Otay Mesa vs Other San Diego Areas:

	comparison_area	count	mean	median	min	max
0	Otay Mesa	493	0.320568	0.275239	0.006106	1.188919
1	Other San Diego Areas	2399	0.840716	0.554101	0.001640	4.753636

Top planning areas by business count:

	planning_area	count	mean	median	min	max
29	OTAY MESA	406	0.321972	0.308361	0.023580	1.083584
31	Outside City Planning Areas	392	0.894668	0.648828	0.001640	4.753636
21	MIRA MESA	369	1.599165	1.725314	0.041083	2.788775

12	KEARNY MESA	206	0.308139	0.269607	0.021394	0.936455
5	CLAIREMONT MESA	108	0.679485	0.752636	0.028897	1.622381
25	NAVAJO	96	1.066008	1.088923	0.021671	2.378257
30	OTAY MESA-NESTOR	87	0.314014	0.265166	0.006106	1.188919
51	UNIVERSITY	78	0.593928	0.559410	0.082059	1.312706
8	DOWNTOWN	74	0.342543	0.367997	0.055863	0.830177
15	MID-CITY:CITY HEIGHTS	68	0.615389	0.525621	0.038312	1.668407

Exported: industrial_businesses_broad_distance_to_freight_roads.csv

Overall summary:

	definition	total_businesses	mean_distance_mi	median_distance_mi	\
0	Broad	4082	0.740517	0.465972	

	min_distance_mi	max_distance_mi
0	0.00164	5.324638

Otay Mesa vs Other San Diego Areas:

	comparison_area	count	mean	median	min	max
0	Otay Mesa	550	0.319213	0.270999	0.005539	1.226746
1	Other San Diego Areas	3532	0.806123	0.543509	0.001640	5.324638

Top planning areas by business count:

	planning_area	count	mean	median	min	max
31	Outside City Planning Areas	595	0.843485	0.607317	0.001640	5.324638
29	OTAY MESA	435	0.320223	0.305615	0.023580	1.226746
21	MIRA MESA	430	1.554692	1.658121	0.041083	2.788775
12	KEARNY MESA	217	0.306654	0.266784	0.021394	0.936455
15	MID-CITY:CITY HEIGHTS	200	0.773906	0.719419	0.035263	1.668407
5	CLAIREMONT MESA	156	0.706135	0.791654	0.028897	1.622381
25	NAVAJO	128	1.020952	0.855477	0.021671	2.378257
30	OTAY MESA-NESTOR	115	0.315392	0.265166	0.005539	1.188919
8	DOWNTOWN	113	0.333455	0.280444	0.055863	0.920747
51	UNIVERSITY	101	0.545879	0.520865	0.060993	1.312706

Analysis parameters:

CRS: EPSG:3310

Buffer distances (m): {'buffer_quarter_mi': 402, 'buffer_half_mi': 805, 'buffer_one_mi': 1609}

Full analysis: run Code_Joseph.ipynb and Code_canyu_June_7th.ipynb

[]: